Cycles identifying vertices for fault isolation in locally twisted cubes *

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Abstract

A set of subgraphs $C_1, C_2, \ldots, C_k$ in a graph G is said to identify the vertices if the sets $\{j \mid v \in C_j\}$ are nonempty for all the vertices v and no two are the same set. We consider the problem of minimizing k when the subgraphs C_i are required to be cycles. The motivation comes from fault diagnosis of multiprocessor systems. We study the cases when G is the locally twisted cube.

1 Introduction

Rapid advances in semiconductor technology have made possible the design of systems with a large number of components. It is difficult to build such systems without defects. Thus fault diagnosis have become issues of great interest in design and analysis of computer. It was in this context that Preparata, Metze and Chien [17] proposed a model, now well known as the PMC model, and a framework for the diagnosis of large scale digital systems. The process of identifying faulty processors is called diagnosis of the system. In this model, every processor performs tests on its neighbors based on the communication links between them. When one processor tests another, the tester declares the tested processor to be fault-free or faulty depending on the test response; the result is always accurate if the tester is fault-free, but if the tester is faulty, the result is unreliable. The collection of all test results is called a syndrome. The problem is to identify all faulty processors subject to certain assumptions. Numerous studies have been dedicated to the PMC model [3, 4, 7, 12, 13, 16, 19, 24].

Karpovsky et al. [8] introduced the concept of identifying codes and studies several issues relating to the construction of these codes. Consider an undirected graph G with vertex set V and edge set E. A ball of radius $t \geq 1$ centered at a vertex v is defined as the set of all vertices that are at distance t or less from v. The vertex v is said to cover itself and all the vertices in the ball with v as the center. The identifying codes problem defined by Karprovsky et al. [8] is to find a minimum set D such that every vertex in G belongs to a unique set of balls of radius $t \geq 1$ centered at the vertices in D. The set D may be viewed as a code identifying the vertices and is called an identifying code. Assume that we want to maintain a system G, and consider the situation in which at lost one of the processors is not working. A lot of work has been done, when the identification of malfunctioning processors is made using balls with v as the center, i.e., all the processors that are within distance r and reports YES/NO depending on whether it has detected a problem or not. Based on these YES/NO answers, we want to be able to tell the exact location of the malfunctioning processor or that all the processors are fine under the assumption that there is at most one malfunctioning processor. Further results for typical multiprocessor architectures have been presented, for example, in [1, 2, 5, 9, 10, 18] for hypercubes.

In [6], the idea of using cycles instead of balls is mentioned. The mathematical problem can then be formulated as follows. We send test messages and can route them through this network in any way we like. What is smallest number of messages we have to send if based on which messages safely come back we can tell which vertex is faulty? We call these the vertex identification problem. The locally twisted cube is a well-known variant of the

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classical hypercube proposed by Yang et al. [22] and has been attracting mush research interest in literatures since its proposal [14, 15, 21, 20, 22]. In this paper, we study the vertex identification problem on the locally twisted cube.

The rest of this paper is organized as follows: In Section 2, a formal description of the locally twisted cube is given and some useful notations are defined, including notation for the permutation of link dimension and the reflected link label sequence. Section 3 presents the sufficient and necessary conditions for determining cycles in LTQ_n based on the reflected link label sequence approach. Using these approaches, main results are shown in Section 4. Finally, we give some conclusions.

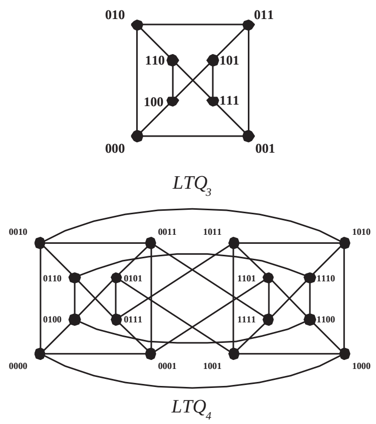

Figure 1: LTQ_3 and LTQ_4

2 Preliminaries

A topology of an interconnection network is conveniently represented by an undirected simple graph $G = (V, E)$, where $V(G)$ and $E(G)$ is the vertex set and the edge set of G, respectively. For graph terminology and notation not defined here we refer the reader to [11]. A walk in a graph is a finite sequence $\omega : \lambda_0, e_1, \lambda_1, e_2, \lambda_2, \ldots, \lambda_{k-1}, e_k, \lambda_k$ whose terms are alternately vertices and edges so, for $1 \le i \le k$, the edge e_i has ends λ_{i-1} and λ_i, thus each edge e_i is immediately preceded and succeeded by the two vertices with which it is incident. In particular, a walk ω is called a path if all internal vertices, λ_i for $1 \le i \le k-1$, of the walk ω are distinct. Both of two vertices λ_0 and λ_k are called end-vertices of the path ω. For simplicity, the path ω is also denoted by $\lambda_0, \lambda_1, \ldots, \lambda_k$. If $\lambda_0 = \lambda_k$, then ω is called a cycle. A cycle of length l is called a l-cycle. A path (respectively, cycle) traversing each vertex of G exactly once is the *Hamiltonian path* (respectively, *Hamiltonian cycle*).

Let $\{0,1\}^n$ denote the set of all binary strings of length n. For two binary strings x and $y \in \{0,1\}^n$, let $x + y$ denote the (bitwise modulo 2) sum of x and y. For every integer $0 \le i \le n-1$, let b_i denote the binary string $x_{n-1}x_{n-2}\ldots x_0$ with $x_i = 1$ and $x_j = 0$ for all $j \ne i$. For every integer $2 \le i \le n-1$, let B_i denote the binary string $x_{n-1}x_{n-2}\ldots x_0$ with $x_i x_{i-1} = 11$ and $x_j = 0$ for all $j \ne i, i-1$. In addition, let $B_1 = b_1$ and $B_0 = b_0$. As a result, $B_i = b_i$ for $i \le 1$ and $B_i = b_i + b_{i-1}$ for $i \ge 2$, moreover, $b_i + b_i = B_i + B_i = 0^n$ where 0^n denote a string consisting of n 0s.

Definition 1 *[22] For $n \ge 2$, an n-dimensional locally twisted cube, denoted by LTQ_n, is defined recursively as follow:*

(1) *LTQ_2 is a graph consisting of four nodes labeled with 00, 01, 10, and 11 respectively, connected by four edges (00, 01),(00, 10),(01, 11) and (10, 11).*

(2) *For $n \ge 3$, LTQ_n is built from two disjoint copies of LTQ_{n-1} according to the following steps. Let $0LTQ_{n-1}$ (respectively, $1LTQ_{n-1}$) denote the graph obtained by prefixing the label of each node in one copy of LTQ_{n-1} with 0 (respectively, 1). Each node $0x_{n-2}x_{n-3}...x_0$ in $0LTQ_{n-1}$ is connected to the node $1(x_{n-2} + x_0)x_{n-3}...x_0$ in $1LTQ_{n-1}$ by an edge.*

Figure 2 shows examples of locally twisted cubes, LTQ_3 and LTQ_4. Either $x = y + b_k$ or $x = y + B_k$ for some $0 \le k \le n-1$ if vertices x and y of LTQ_n are adjacent. Therefore, we call y as the k-neighbor of x and (x, y) is labeled by k; besides, (x, y) is called to be type b if $x = y + b_k$ and type B if $x = y + B_k$.

A path in LTQ_n might be specified by the source vertex and a sequence of labels detailing the edges to be traversed, for example, the path in LTQ_3 detailed as having the source vertex 000 and then following the edges labeled 0-2-1 (also denoted as [0-2-1]) is actually the path 000, 001, 111, 101, also denoted as $000[0-2-1]101$, where $001 = 000 + b_0$, $111 = 001 + B_2$, and $101 = 111 + B_1$. Therefore, the sequence $L = [d_0 - d_1 - \cdots - d_{m-1}]$ is called an *Link Label Sequence* in LTQ_n if two adjacent labels are not identical where $d_i \in Z_n$, $Z_n = \{0, 1, \ldots, n-1\}$, for $0 \le$

$i \le m-1$. A walk, $\omega(L,u) = \lambda_0, \lambda_1, \lambda_2, \ldots, \lambda_m$, in LTQ_n can be generated with respect to a given link label sequence L and a given vertex u as follows: $\lambda_0 = u$, and λ_j is the d_{j-1}-neighbor of λ_{j-1} in LTQ_n where $0 \le j \le m$. Thus, this walk $\omega(L,u)$ is also represented as $\lambda_0[L]\lambda_m$.

An interesting approach to generate such a path in n-dimensional cube is the *reflected link label* sequence (as defined below), which were proposed by Zheng and Latifi [23]. Based on a reflected link label sequence, Zheng and Latifi defined a generalized Gray Code (GGC) and utilized GGCs to identify Hamiltonian paths and cycles in an n-dimensional crossed cube. From now on, we will propose a systematic approach based on reflected link label sequence to construct Hamiltonian cycles in an n-dimensional locally twisted cube.

Definition 2 *[23] For $n \ge 1$, let $\pi = \langle d_1, d_2, \ldots, d_n\rangle$ be a permutation over on Z_n, i.e., $\pi \in \Pi_n$, and $\pi(i) = \langle d_1, d_2, \ldots, d_i\rangle$, for $1 \le i \le n$. Then the reflected link label sequence, denoted as G_π, defined by π can be generated recursively as follows:*

$$G_{\pi(1)} = d_1,$$
$$G_{\pi(k)} = G_{\pi(k-1)} - d_k - G_{\pi(k-1)}, 2 \le k \le n;$$
$$G_\pi = G_{\pi(n)}.$$

As a result, the $G_{\pi(k)}$ is a symmetry link label sequence in LTQ_n and hence, $\omega(G_{\pi(k)}, u)$ comes into being a walk in LTQ_n with a given starting vertex u. Subsequently, the following lemma is easily obtained and thus details of the proof are omitted.

Lemma 1 *Let $\pi = \langle d_1, d_2, \ldots, d_n\rangle \in \Pi_n$. Then the total number of d_i appearing in $G_{\pi(k)}$ is 2^{k-i}, where $1 \le i \le k$.*

In addition, the following result could be concluded according to Lemma 1.

Lemma 2 *Let $\pi = \langle d_1, d_2, \ldots, d_n\rangle \in \Pi_n$. Then, the total number of d_i appearing between any two successive d_j's in $G_{\pi(k)}$ is 2^{j-i} if $1 \le i < j \le k-1$.*

Proof. Let $k \le n$ and $0 \le i < j \le k-1$. Hence $\pi(j+1) = \langle d_1, d_2, \ldots, d_i, \ldots, d_{j+1}\rangle$. Recall that $G_{\pi(j+1)} = G_{\pi(j-1)} - d_j - G_{\pi(j-1)} - d_{j+1} - G_{\pi(j-1)} - d_j - G_{\pi(j-1)}$. Since $G_{\pi(j+1)}$ is a subsequence of $G_{\pi(k)}$ if $j \le k-1$, the total number of d_i appearing between any two successive d_j's in $G_{\pi(k)}$ is equal to the total number d_i of appearing between two d_j's in $G_{\pi(j+1)}$. By Lemma 1, the total number of d_i appearing in $G_{\pi(j-1)}$ is 2^{j-1-i}. Therefore, there are $2 \cdot 2^{(j-i-1)} = 2^{(j-i)}$ d_i's between any two successive d_j's in $G_{\pi(k)}$, for $0 \le i < j \le k-1$. □

Before our further works, the following definition and lemmas are necessary. For any two vertices, $u = u_{n-1}u_{n-2}\ldots u_1u_0$ and $v = v_{n-1}v_{n-2}\ldots v_1v_0$, in LTQ_n, define $Diff(u,v) = \{i \mid u_i \ne v_i \text{ for } 0 \le i \le n-1\}$. For instance, $Diff(0001, 1010) = \{0,1,3\}$. Therefore, we can immediately obtain

Lemma 3 *For any link label sequence L, let $u[L]w$ and $v[L]z$ be two walks in LTQ_n. Then, $Diff(u,v) = Diff(w,z)$ if $u_0 = v_0$.*

3 Reflected link label sequence and Cycle

In this section, we firstly show the sufficient and necessary conditions for determining Hamiltonian paths and cycles in LTQ_n based on the reflected link label sequence approach.

Lemma 4 *For $u \in V(LTQ_n)$ and $\pi \in \Pi_n$, the walk $\omega(G_{\pi(k)}, u)$ is a path passing through 2^k vertices in LTQ_n.*

Proof. Let $\pi = \langle d_1, d_2, \ldots, d_n\rangle \in \Pi_n$. And hence $\pi(k) = \langle d_1, d_2, \ldots, d_k\rangle$. We will claim $\omega(G_{\pi(k)}, u)$ is a path with 2^k vertices in LTQ_n for any starting vertex u, i.e., any two distinct vertices lying on $\omega(G_{\pi(k)}, u)$ are not identical. We argue by induction on k. For $k = 1$, the permutation $\pi(1) = \langle d_1\rangle$ is considering. Clearly, for any vertex $u \in V(LTQ_n)$, $\omega(G_{\pi(1)}, u)$ is an edge in LTQ_n. Assume that for $k \le m-1$, $2 \le m < n$, the lemma is true.

Let $k = m$. By definition, we have $G_{\pi(m)} = G_{\pi(m-1)} - d_m - G_{\pi(m-1)}$. Thus $\omega(G_{\pi(m)}, u)$ can be rewrite as $u[G_{\pi(m-1)}]x[d_m]y[G_{\pi(m-1)}]v$. One can observe that $\omega(G_{\pi(m-1)}, x) = x[G_{\pi(m-1)}]u$ and $\omega(G_{\pi(m-1)}, y) = y[G_{\pi(m-1)}]v$. By the induction hypothesis, it is obvious that $\omega(G_{\pi(m-1)}, x)$ and $\omega(G_{\pi(m-1)}, y)$ are paths with 2^{m-1} vertices respectively. Let w be any vertex lying on the path $x[G_{\pi(m-1)}]u$ and z be any vertex lying on the path $y[G_{\pi(m-1)}]v$, respectively. We now claim $w \ne z$.

The lemma is obvious if $d_m = 0$ and $n-1$. Thus, we only consider $d_m \ne 0, n-1$. Let w be an end-vertex of the path $x[L_1]w$ and z be an end-vertex of the path $y[L_2]z$, where L_1 and L_2 are two sub-sequences of $G_{\pi(m-1)}$. By Lemma 3,

$Diff(x,y) = Diff(w,z)$ if $L_1 = L_2$. This implies $w \neq z$ if $L_1 = L_2$. From now on, we consider the case of $L_1 \neq L_2$. Without loss of generality, assume the length of L_1 is smaller than that of L_2. Let $y[L_2]z$ be decomposed as $y[L_1]w'[L_3]z$. By Lemma 3, $Diff(x,y) = Diff(w',w)$ due to $x_0 = y_0$. Note, $Diff(x,y) = Diff(w',z)$ if $z = w$. This implies $z \neq w$ if $Diff(x,y) \neq Diff(w',z)$.

We now claim $Diff(x,y) \neq Diff(w',z)$. Suppose that $Diff(x,y) = Diff(w',z)$. One can observe that $Diff(x,y) \neq Diff(w',z)$ if there are odd number of appearances of 0 in L_3. Thus, L_3 contains even numbers of 0. Since x is the d_m-neighbour of y, $d_m \in Diff(x,y)$. Recall d_m is not appearing in $\pi(m-1)$. This implies only the element d_m+1 in $\pi(m-1)$ can change the value of the d_m-th bit position of w' and z. Let $d_i = d_m+1$ and $d_j = 0$ for some $1 \leq i,j \leq m-1$. Let s denote the total number of d_i in the sequence L_3. There are two possibilities with respect to the relationship between i and j.

Case 1: $i < j$. By Lemma 2, there are even numbers of d_m+1 between two successive 0's in $G_{\pi(m-1)}$. Since L_3 contains even numbers of 0, all edges with label d_m+1 appearing in the path $w'[L_3]z$ are partitioned into two subsets that one contains even numbers of one type, b-type or B-type, edges and the other contains remaining edges belonging to the other type. Since $b_{d_m} + b_{d_m} = 0^n$ and $B_{d_m} + B_{d_m} = 0^n$, $d_m \notin Diff(w',z)$ if s is an even integer; otherwise, $\{d_m+1\} \subseteq Diff(w',z)$ or $\{d_m, d_m+1\} \subseteq Diff(w',z)$. Since $Diff(x,y) = \{d_m\}$ or $Diff(x,y) = \{d_m, d_m-1\}$, $Diff(x,y) \neq Diff(w',z)$. A contradiction occurs.

Case 2: $i > j$. By Lemma 2, there are even numbers of 0 between two successive (d_m+1)'s in $G_{\pi(m-1)}$. All edges with label d_m+1 appearing in the path $w'[L_3]z$ are in the same type. The argument is similar to that of Case 1. This completes our inductive proof. □

Theorem 1 *For $n \geq 2$, let $u \in V(LTQ_n)$ and $\pi = \langle d_1, d_2, ..., d_n\rangle \in \Pi_n$. Then, the walk $\omega([G_\pi - d_n], u)$, generated by the reflected link label sequence G_{π_n} and the starting vertex u in LTQ_n, corresponds to a Hamiltonian cycle of LTQ_n if and only if one of the following conditions holds:*

(1) $d_{n-1} \cdot d_n \neq 0$, *or*

(2) $|d_{n-1} - d_n| = 1$.

Proof. $(\Leftarrow)$
Case 1: $d_{n-1} \cdot d_n \neq 0$, i.e., $d_n \neq 0$ and $d_{n-1} \neq 0$. Since $G_\pi = G_{\pi(n-1)} - d_n - G_{\pi(n-1)}$, let $\omega(G_\pi, u) = u[G_{\pi(n-1)}]x[d_n]y[G_{\pi(n-1)}]v$, where $\omega(G_{\pi(n-1)}, x) = x[G_{\pi(n-1)}]u$ and $\omega(G_{\pi(n-1)}, y) = y[G_{\pi(n-1)}]v$. Since $d_n \neq 0$, we have $x_0 = y_0$, where $x = x_{n-1}x_{n-2}\ldots x_1x_0$ and $y = y_{n-1}y_{n-2}\ldots y_1y_0$. Thus, by Lemma 3, $Diff(x,y) = Diff(u,v)$. Further, by Lemma 1, $d_{n-1} \neq 0$ implies the total number of appearances of 0 in $G_{\pi(n-1)}$ is even, which implies $x_0 = y_0 = u_0 = v_0$ where $u = u_{n-1}u_{n-2}\ldots u_1u_0$ and $v = v_{n-1}v_{n-2}\ldots v_1v_0$. Therefore, (u,v) is an edge with label d_n. Hence the path $\omega([G_\pi - d_n], u)$ corresponds to a Hamiltonian cycle of LTQ_n.

Case 2: $|d_{n-1} - d_n| = 1$. In this case, we need only to verify cases of that $d_{n-1} = j$ and $d_n = 1-j$ for $j = 0,1$. Suppose $d_{n-1} = 1$ and $d_n = 0$. Let $\omega(G_\pi, u) = u[G_{\pi(n-1)}]x[d_n]y[G_{\pi(n-1)}]v$. Since $d_n = 0$, we have $x_0 = \overline{y_0}$. Besides, $u_1 = \overline{x_1}$ and $v_1 = \overline{y_1}$ because of $d_{n-1} = 1$. By Lemma 1, the number of d_i appearing in the sequence $G_{\pi(n-1)}$ is even for all $1 \leq i \leq n-2$ and d_{n-1} appears once in $G_{\pi(n-1)}$. Hence, $Diff(u,v) = \{0\}$, which implies (u,v) is an edge with label 0. Thus, the path $\omega([G_\pi - d_n], u)$ corresponds to a Hamiltonian cycle of LTQ_n. If $d_{n-1} = 0$ and $d_n = 1$, the argument is similar.

$(\Rightarrow)$ We argue by refutation. If $d_{n-1} \cdot d_n = 0$ and $|d_{n-1} - d_n| > 1$, we have that $d_{n-1} = k$ and $d_n = 0$, or $d_{n-1} = 0$ and $d_n = k$ for some $k \geq 2$. Suppose $d_{n-1} = k$ and $d_n = 0$ for some $k \geq 2$. Let $\omega(G_\pi, u) = u[G_{\pi(n-1)}]x[d_n]y[G_{\pi(n-1)}]v$, where $\omega(G_{\pi(n-1)}, x) = x[G_{\pi(n-1)}]u$ and $\omega(G_{\pi(n-1)}, y) = y[G_{\pi(n-1)}]v$. We have $Diff(x,y) = \{0\}$ due to $d_n = 0$. Note, $\pi(n-1)$ is a permutation over on $Z_n - \{0\}$. Then, there is no 0 in $G_{\pi(n-1)}$. Hence, $0 \in Diff(u,v)$. By definition $G_{\pi(n-1)}$, the total number of k, appearing in $G_{\pi(n-1)}$ is only one. Therefore $k-1 \in Diff(u,v)$. Thus, since $k \geq 1$, u is not a neighbour of v. A contradiction occurs. If $d_{n-1} = 0$ and $d_n = k$ for some $k \geq 2$, the argument is similar. This completes the proof of the theorem. □

Following two lemmas are useful and straightforward.

Lemma 5 *For $n \geq 2$, let $u \in V(LTQ_n)$ and $\pi = \langle d_1, d_2\rangle$. Then, the walk $\omega([G_\pi - d_n], u)$ is a 4-cycle if and only if one of the following conditions holds:*

(1) $d_{n-1} \cdot d_n \neq 0$, *or*

(2) $|d_{n-1} - d_n| = 1$.

Lemma 6 *Let $u \in V(LTQ_n)$ and $\pi = \langle d_1, d_2, ..., d_m\rangle$ where $0 \leq d_i \leq n-1$. the walk*

$\omega(G_\pi, u)$ is a cycle of length 2^m vertices in LTQ_n if the following conditions holds:

(1) $d_{m-1} \cdot d_m \neq 0$, *or*

(2) $|d_{m-1} - d_m| = 1$.

4 Main Results

In order to solve vertex identification problem on the locally twisted cube LTQ_n, we wish to find a collection of cycles $C_1, C_2, \ldots, C_k$ such that every vertex belongs to at least one of them and, moreover, the sets $\{j \mid v \in C_j\}$ are all pairwise different. We denote the minimum cardinality k for LTQ_n by $ID_v(n)$. For arbitrary sets, we have following trivial identification theorem:

Theorem 2 *A collection $A_1, A_2, \ldots, A_k$ of subsets of an s-element set S is called identifying, if for all $x \in S$ the sets $\{i \mid x \in A_i\}$ are nonempty and different. Given s, the smallest identify collection of subsets consists of $\lceil log_2(s+1) \rceil$ subsets.*

The vertex identification problem is special cases of this problem, and for the locally twisted cube LTQ_n with $s = 2^n$ vertices we get the lower bound $ID_v(n) \geq \lceil log_2(2^n+1) \rceil = n+1$.

Let $\pi = \langle d_1, d_2, \ldots, d_m \rangle$ where $0 \leq d_i \leq n-1$. For simplicity, we use $\Omega(\pi, u)$ to denoted the walk $\omega([G_\pi - d_m], u)$ if $\omega([G_\pi - d_m], u)$ is a cycle in LTQ_n. Let 0^n and 1^n denote all-zero and all-one n-bit binary strings, respectively.

For $n \geq 4$, let $\pi^i = \langle 0, 1, 2, \ldots, i-1, i+1, i+2, \ldots, n-1 \rangle$ where $i \geq 2$. By Lemma 2, we have that the total number of 0 appearing between two successive $i+1$ in G_{π^i} is 2^{i-1} and total number of 0 appearing before $i+1$ is 2^{i-1} too. Since G_{π^i} contains none of i, the path $\omega(G_{\pi^i}, u)$ starting any vertex u consists of 2^{n-1} vertices whose ith coordinate equals u_i. Therefore, we have following useful Lemmas.

Lemma 7 *For $n \geq 4$, let $\pi^i = \langle 0, 1, 2, \ldots, i-1, i+1, i+2, \ldots, n-1 \rangle$. Then, the cycle $\Omega(\pi^i, 0^n)$ passes all vertices v with $v_i = 0$ where $v = v_{n-1}v_{n-2}\ldots v_0$ for $i \geq 2$.*

Similarly,

Lemma 8 *For $n \geq 4$, let $\pi = \langle 0, n-1, 2, 3, \ldots, n-2 \rangle$. Then, the cycle $\Omega(\pi, 0^n)$ passes all vertices v with $v_1 = 0$ where $v = v_{n-1}v_{n-2}\ldots v_0$.*

Theorem 3 *$ID_v(n) = n+1$ for all $n \geq 3$*

Proof. The proof is divided into two cases: (1) $n = 3$ and (2) $n \geq 4$.

Case 1. $n = 3$. Let

$$\begin{aligned} C_0 &= 000, 100, 110, 010, 000 \\ C_1 &= 000, 100, 101, 011, 001, 000 \\ C_2 &= 000, 001, 011, 010, 000 \\ C_3 &= 100, 101, 111, 110, 000 \end{aligned}$$

Obviously, every vertex of LTQ_3 belongs to at least one of the cycles C_0, C_1, C_2, C_3 and moreover, the sets $\{j \mid v \in C_j\}$ are all pair distinct.

Case 2. $n \geq 4$. We construct $n+1$ cycles all starting from the all-zero binary string 0^n. Let $\pi = \langle 0, 1, 2, \ldots, n-1 \rangle$. Let $C_n = \Omega(\pi, 0^n)$. By Theorem 1, C_n is a Hamiltonian cycle of LTQ_n. The n cycles $C_0, C_1, \ldots, C_{n-1}$ are constructed as follows.

Let $C_0 = \Omega(\pi^0, 0^n)$ and $C_1 = \Omega(\pi^1, 0^n)$ where $\pi^0 = \langle 1, 2, \ldots, n-1 \rangle$ and $\pi^1 = \langle 0, n-1, 2, 3, \ldots, n-2 \rangle$, respectively. It is observed that C_0 is a cycle of length 2^{n-1} which passes all vertices whose 0th coordinate equals 0. Similarly, by Lemma 8, C_1 is a cycle consists of all vertices whose 1th coordinate equals 0. Given $2 \leq i \leq n-1$, $\pi^i = \langle 0, 1, 2, \ldots, i-1, i+1, i+2, \ldots, n-1 \rangle$. Let $C_i = \Omega(\pi^i, 0^n)$. By Lemma 7, C_i is a cycle of length 2^{n-1} which visits exactly once all the 2^{n-1} vertices whose ith coordinate equals 0.

Therefore, a vertex v lies in C_i if and only if $v_i = 0$. The cycle C_n guarantees that also the 1^n vertex lies in at least one cycle. □

5 Conclusions

System level diagnosis proposed in [17] and the fault diagnosis framework based on identifying codes [8] are two powerful approaches in fault diagnosis of computer and communication systems. There are two main differences in the application of the above two frameworks in fault diagnosis. In the application of the system level diagnosis we are required tto construct a test graph that satisfies certain measures of diagnosis as desired by the application. Constructing such a graph is not always easy. Also, we often need sophisticated algorithms for diagnosis using the test graph. In applying the identifying codes technique to diagnosis, we need to find a identifying code of the given graph. In certain cases an identifying code may not even exist. If such a code does not exist, we can decompose the graph into simple subgraphs for which identifying codes exist. Then we can place monitors on each of the subgraphs and use them to isolate faults within those subgraphs. In this paper, identifying codes can be constructed by cycle for locally twisted cubes. We solve the vertex identification problem on locally twisted cubes.

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